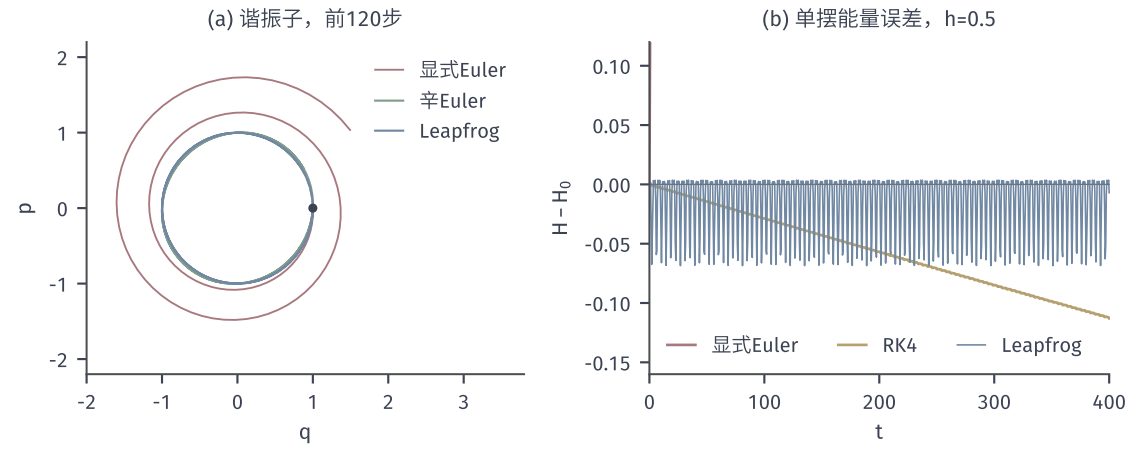
(a) 谐振子，前120步
显式Euler
辛Euler
Leapfrog
p
q
2
1
0
-1
-2
-2
-1
0
1
2
3
(b) 单摆能量误差，h=0.5
H – H0
0.10
0.05
0.00
-0.05
-0.10
-0.15
显式Euler
RK4
Leapfrog
t
0
100
200
300
400